RESEARCH IN SCIENCE & TECHNOLOGICAL EDUCATION

01/03

NEW ARTICLE

Comparing the effectiveness of Science Writing Heuristic and Model-based Science Writing Heuristic laboratory activities on problem-solving perceptions of pre-service elementary teachers

Nejla Atabey

Muş Alparslan University, Türkiye

DOI **doi.org/10.1080/02635143.2026.2742294**

Taylor & Francis
Taylor & Francis Group

WHAT DID THE RESEARCHERS INVESTIGATE?

- Traditional, recipe-like laboratory practices have been criticised for offering limited cognitive engagement.
- The study compared the Science Writing Heuristic (SWH) with the Model-Based Science Writing Heuristic (M-SWH), which integrates modelling with argumentation and writing.
- A quasi-experimental pre-test–post-test design measured the problem-solving perceptions of 53 pre-service elementary teachers.

Atabey (2026) · Research in Science & Technological Education · doi.org/10.1080/02635143.2026.2742294

WHAT DID THEY FIND?

- Both approaches had significant effects with large effect sizes on the teachers' problem-solving perceptions.
- The M-SWH approach improved problem-solving perceptions significantly more than the SWH-based approach.
- The study may support integrating lab activities that combine argumentation, modelling and writing into undergraduate programmes.

Atabey (2026) · Research in Science & Technological Education · doi.org/10.1080/02635143.2026.2742294